

SAP LABS FTE INTERVIEW EXPERIENCE

- Shilpi Gupta

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Round 1: Technical Round 1 (55 minutes)

- Introduce yourself.
- Asked me to explain my ML project (The interviewer cross questioned me 3-4 times with WHATs and WHYs and the discussion lasted for 10 minutes).
- Coding question: Print top, left view of the binary tree.
<https://www.geeksforgeeks.org/print-left-view-binary-tree/>
<https://www.geeksforgeeks.org/print-nodes-top-view-binary-tree/>
- Difference between array/linked list.
- Difference between malloc, calloc. (Deep discussion).
- Difference between C-C++, Procedural – Object Oriented.
- A C++ program was given to me with final keyword and I was supposed to tell the output of the program with the reason.
Answer: Program will give error.
- Does JAVA support multiple inheritance? If yes, Why?
- What is friend function? Explain with example.
- What is abstraction and why is it used?(write example code)
- Difference between abstract class and interface.
- Difference between Merge Sort and Quick Sort (Compare with respect to their time complexities).
- Real life example of LIFO/FIFO.
- Difference between multithreading and paging. Also asked continued questions on the answers I was giving).
- What is IPC and it's contribution in Distributed systems?
- Explain deadlock and it's resolution.
- Explain the whole process of segmentation.
- **SQL:** Two tables were given:
Table 1: Student_Id, Student_Name
Table 2: Subject_Id, Student_Name, Student_Id

- Table 2 contained the IDs of the students who had taken up the particular subjects.
 - In Table 2, Student_id has some NULL values.
 - A nested SQL query was given, and I was supposed to tell the output.
- What is Indexing and their types?
 - Why indexing is used in DBMS?
 - What is the difference between AI and Machine Learning?
 - Any questions to ask? (I asked two questions from him)
 - How was your journey from college to SAP Labs?
 - Any suggestions? As I am going to start my career in the IT sector.
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Round 2: Technical Round 2 (1 hour)

This round was totally based on judging my analytical thinking and how I'm approaching to a given problem.

- Introduce yourself.
- Tell me about your journey from school to MCA.
- Why you chose MCA after BSc.(Mathematics)? Why not higher studies in Mathematics? Why chose technology over Mathematics?
- The interviewer had my resume. He asked me to explain both of my projects. (He was diving into and picking points from whatever I was telling about the project and so cross questioning),(15-20 min).
- Which is your favourite competitive programming platform?
He started with the follow up questions-
 - Suppose there is a company just like any competitive programming platform, who do you think will be the target audiences of the company? (I answered him by giving two to three points).
- He asked me to share my screen and open Google Chrome Home page.
 - He asked me to consider myself as a software developer and tell how am I going to test the Google search engine? Where it can be failed for the users?

- He asked me to look at each and every icon on the page. And how are you going to test each icon as a developer?
- For example- when a user is voice searching, how is the speed of the search? The clarity of voice and detection by the page. Is it able to catch the voice and search by the right words spoken?
- We had a lengthy discussion at each step, he wanted me to expand my answer domain.
- Started with another question-
There is a question of finding the prime numbers, the input is taken in a 'TEXT BOX'. Explain each and every test case and use cases related to this problem also considering the cases of possible inputs in a text box. (As a problem setter).
- As you have done a machine learning project, So suppose I am a 10 year old kid, how are you going to explain Machine Learning to me. Start Explaining.
- From my resume he asked me- How I taught Hindi to the South Indians at NIT Trichy ? I told him the way I used to explain Hindi by referring and comparing to the English language. Again he cross questioned- How were you able to explain gender difference of objects like in Hindi few objects are considered to be feminine and few to be masculine but that is not the case with the English language.
- Asked simple coding questions-
 - How are you going to find missing and repeating number in a given array? (Kept on changing the problem and asking for solutions- asked to write the function only)
<https://www.geeksforgeeks.org/find-a-repeating-and-a-missing-number/>
 - Merge two sorted linked list(asked me to write the code)
<https://www.geeksforgeeks.org/merge-two-sorted-linked-lists/>
- Puzzle:
You're given three boxes with red balls in the first one, blue in the second and red and blue both mixed in the third. The boxes are labelled wrongly. Now your task is to correctly label each box with it's corresponding content. How can you do that by opening minimum possible boxes?

- Any questions?(I asked two questions)

P.S. - At the end of the interview, the interviewer complemented me for my comprehensive explanations to the answers and wished me luck for the further rounds.

Round 3: Managerial Round (45 minutes)

- Although this round was managerial technically, but the interviewer only focused on the two projects I mentioned in my Resume.
 - He started with the Machine Learning Project.
 - He asked each and every detail about my project (Example- Asked about the parameters I used and Why? Explain the algorithms used, How the accuracy can differ for different parameters and so many questions).
 - He even opened the dataset on his PC(as it was available on Kaggle) and started asking questions relating to that.
 - Why do you think, different firms are easily leaking their sensitive data?
 - As Random forest algorithm was used in my project, he asked me to share my screen with whiteboard and asked me to take sample data from the dataset and make decision trees from that and show the output.
 - Other follow up questions.
 - Then he started with my second project based on Blockchain. Again had a detailed discussion with many cross questions.
 - Asked me to share the screen and design the system design of my project.
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Round 4: HR Round (20 minutes)

- Brief yourself.
- How you came into MCA from Mathematics?
- Explain any of your projects from starting to end, challenges faced during the making, How you came to the solution?
- Is your project complete or still in progress?
- Are you available to reallocate to Bengaluru?
- Any questions?

Verdict: Selected

Suggestions:

- Prepare your projects well. I was never fully prepared with my projects but in all the four rounds of the interviews I was asked about the project :D
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